🚀 8-Week Pre-Semester Roadmap (Aug 11 – Oct 2026)

Goal: Refresh PyTorch, deepen Multimodal & Generative AI fundamentals, and build research-ready habits before Brac University classes start.

You've already covered a full year of ML/DL/NLP/CV/LLM basics. This plan assumes that foundation exists but is **rusty**, so Weeks 1–2 are a fast, hands-on refresher — not a re-do from scratch. From Week 3 onward, everything is aimed squarely at **multimodal + generative AI**, since that's your research direction.

Week 1: PyTorch Refresher — Core Mechanics

Goal: Get your hands back on the keyboard, not just theory.

- [ ] Rebuild a CNN classifier on CIFAR-10 from memory (no copy-paste)
- [ ] Review `autograd`, custom `nn.Module`, custom loss functions
- [ ] Refresh: DataLoader, Dataset, transforms, checkpointing
- [ ] Learn/relearn `torch.compile()`, mixed precision (`torch.autocast`, `GradScaler`)
- [ ] Set up a clean dev environment: `uv`/`conda`, PyTorch 2.x, CUDA check, Jupyter or VS Code

Deliverable: A single clean repo with a working, trained CNN + training loop you understand line by line.

Week 2: Modern Training Practices

Goal: Bridge the gap between "toy scripts" and how models are actually trained today.

- [ ] Learn efficient fine-tuning: **LoRA** / **QLoRA** (concept + `peft` library)
- [ ] Learn quantization basics (bitsandbytes, 8-bit/4-bit inference)
- [ ] Learn distributed/parallel training concepts: DDP, FSDP (just enough to read code, not necessarily run multi-GPU)
- [ ] Experiment tracking: set up **Weights & Biases** or MLflow on a real run
- [ ] Refresh transformer internals: self-attention, positional encodings, from-scratch mini-GPT

Deliverable: Fine-tune a small open model (e.g. a small BERT/GPT2-class model) with LoRA and log the run in W&B.

Week 3: Vision-Language Foundations

Goal: Understand how vision and text get mapped into shared space.

- [ ] Deep-dive **CLIP**: contrastive loss, image/text encoders, zero-shot classification
- [ ] Implement CLIP-based image search / zero-shot classifier from scratch (using `open_clip` or HF)

- [ ] Study **BLIP** / **BLIP-2** or similar vision-language captioning architectures
- [ ] Read 2–3 recent (2025–2026) multimodal papers from arXiv/Papers with Code

Deliverable: A working "search images by text" mini-tool using CLIP embeddings.

Week 4: Vision-Language Models, Continued

Goal: Move from encoders to generation/reasoning over multiple modalities.

- [ ] Study Vision-Language Models (VLMs) that do captioning/VQA (e.g. LLaVA-style architectures — vision encoder + LLM adapter)
- [ ] Try running/fine-tuning a small open VLM locally or via HF Spaces
- [ ] Learn image-text dataset formats (WebDataset, HF `datasets` for multimodal)
- [ ] Study evaluation metrics for multimodal tasks (CLIPScore, BLEU/CIDEr for captioning)

Deliverable: Run inference with an open VLM on your own images and document outputs/failure cases.

Week 5: Diffusion & Generative Models — Theory

Goal: Understand the math and architecture behind modern image generation.

- [ ] Learn diffusion fundamentals: forward/reverse process, denoising, DDPM vs DDIM
- [ ] Study **Latent Diffusion** (why Stable Diffusion operates in latent space)
- [ ] Study U-Net architecture used in diffusion backbones
- [ ] Understand classifier-free guidance and text conditioning (cross-attention with text embeddings)

Deliverable: Write (or annotate) a from-scratch simplified diffusion training loop on a toy dataset (e.g. MNIST) to internalize the math.

Week 6: Diffusion & Generative Models — Practice

Goal: Move from theory to using and adapting real generative models.

- [ ] Use HF `diffusers` library: run inference, understand pipeline components (VAE, UNet, scheduler, text encoder)
- [ ] Try fine-tuning techniques: **DreamBooth**, **LoRA for diffusion models**, textual inversion
- [ ] Explore newer/faster generative architectures (search for current 2026 state-of-the-art — this space moves fast)
- [ ] Compare diffusion vs. newer approaches (e.g. flow-matching / consistency models) at a conceptual level

Deliverable: Fine-tune a small diffusion model (LoRA-based) on a custom small image set.

Week 7: Research Skills & Tooling

Goal: Build the habits that make you effective in a lab from day one.

- [ ] Practice reading papers systematically (abstract → figures → method → results → limitations)
- [ ] Reproduce one small paper's core result from Papers with Code
- [ ] Learn to navigate arXiv, HF Papers, and relevant conferences (NeurIPS, CVPR, ICML, ICLR) for your subfield
- [ ] Git/GitHub hygiene for research code: branches, reproducibility, `requirements.txt`/`pyproject.toml`, README with results
- [ ] Learn to write a clean experiment log (what you tried, what changed, what happened)

Deliverable: A GitHub repo of your Week 1–6 work, with a proper README, that you'd be comfortable showing a professor.

Week 8: Capstone Mini-Project + Research Prep

Goal: Simulate an actual small research project, end to end.

- [ ] Pick one small, scoped project combining vision + text + generation, e.g.:
  - A CLIP-guided image retrieval tool with a simple UI
  - A LoRA-fine-tuned diffusion model on a themed dataset with a short "results" writeup
  - A small VLM fine-tuned for a niche captioning task
- [ ] Build it end-to-end: data → training/fine-tuning → evaluation → results writeup
- [ ] Write a short report (1–2 pages) as if presenting to a research advisor: motivation, method, results, limitations, next steps
- [ ] Skim your department/lab's recent publications (if known) so you can speak their language on day one

Deliverable: One polished capstone project + writeup, ready to show in your first week of classes.

🔗 Core Resources

- **PyTorch:** official tutorials (pytorch.org), Zero to Mastery PyTorch (Daniel Bourke) as refresher
- **Multimodal/CLIP:** OpenAI CLIP paper + `open_clip` repo, HF multimodal course
- **VLMs:** LLaVA paper/repo, HF Transformers docs for VLMs
- **Diffusion:** "Understanding Diffusion Models" blog-style explainers, HF `diffusers` docs, Stable Diffusion / Flux papers
- **Research skills:** Papers with Code, Two Minute Papers, arXiv Sanity, HF Papers page
- **Tracking/Tooling:** Weights & Biases docs, `peft` (LoRA) docs, `bitsandbytes` docs

> ⚠️ Note: multimodal/generative AI moves fast. Before Week 5–6, do a quick search for "state of the art image generation 2026" and "best open VLMs 2026" to make sure you're training on architectures that are still current, not just what was current in the source papers.

Weekly Time Budget (suggested)

Days	Focus
Mon–Fri (~1.5–2 hrs/day)	Concept study + guided implementation
Weekend (block of 3–4 hrs)	Deliverable / hands-on project work

Total: ~8 weeks, finishing late September/early October 2026 — right as classes start.